

Logistic Population Growth



The equation for logistic growth in discrete time

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The definition of density-dependent growth

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Basic properties of the model

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Basic properties of the model

Strange behavior of the (discrete time) model, such as damped oscillations and chaos

Geometric growth

$$N_{t+1} = N_t + N_t r$$

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Logistic growth

$$N_{t+1} = N_t + N_t r_{max} \left(1 - \frac{N_t}{K} \right)$$

where

- r_{max} is the growth rate when N_t is close to 0.
- K is the carrying capacity

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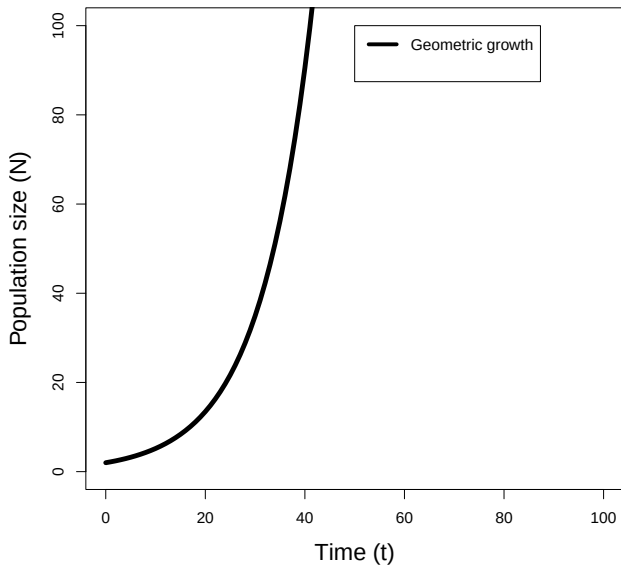
Definition: Population growth rate *is* affected by population size (N).

Logistic growth is an example of **density-dependent growth**

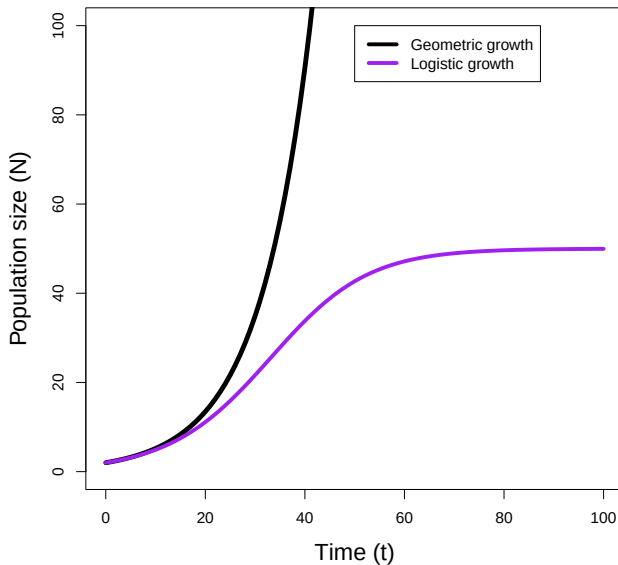
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Implications: Resources are limited and there is a carrying capacity.

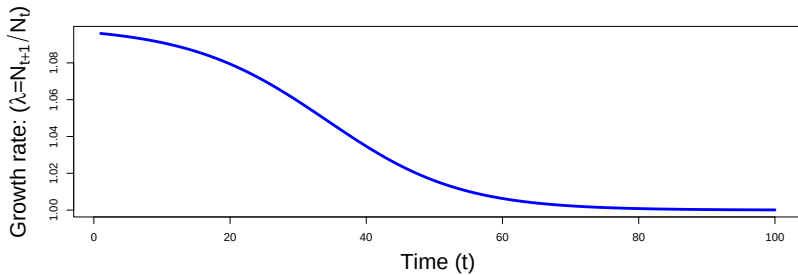
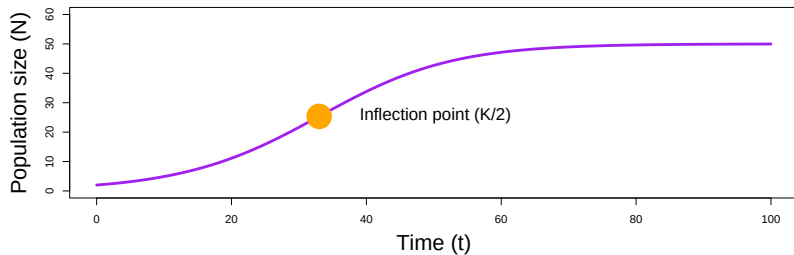
GRAPHICAL DEPICTION



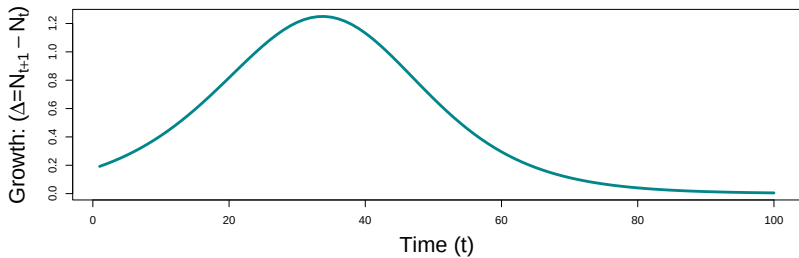
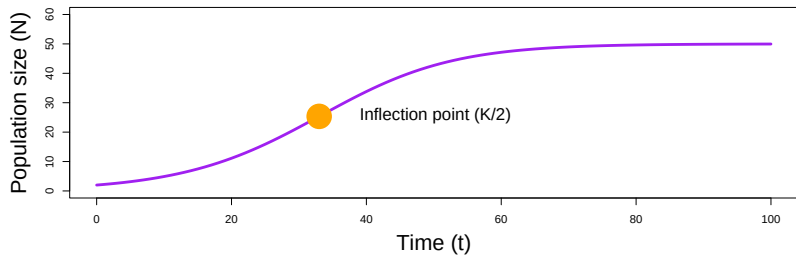
GRAPHICAL DEPICTION



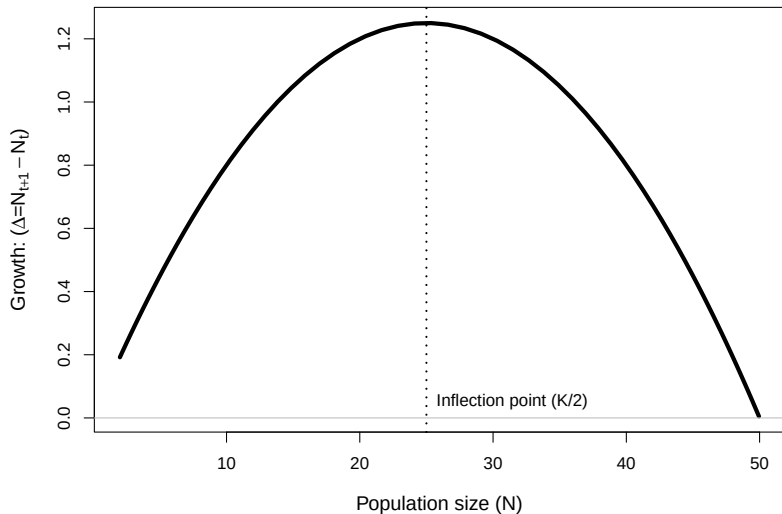
GROWTH RATE ($\lambda_t = N_{t+1}/N_t$)



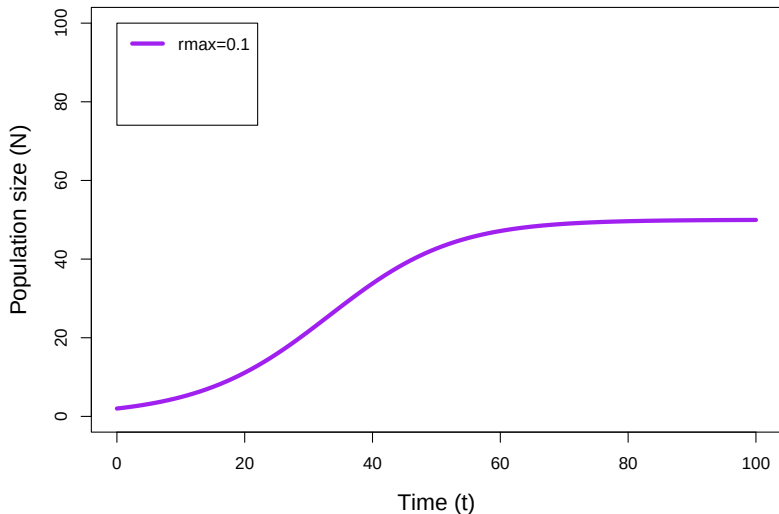
GROWTH ($\Delta_t = N_{t+1} - N_t$)



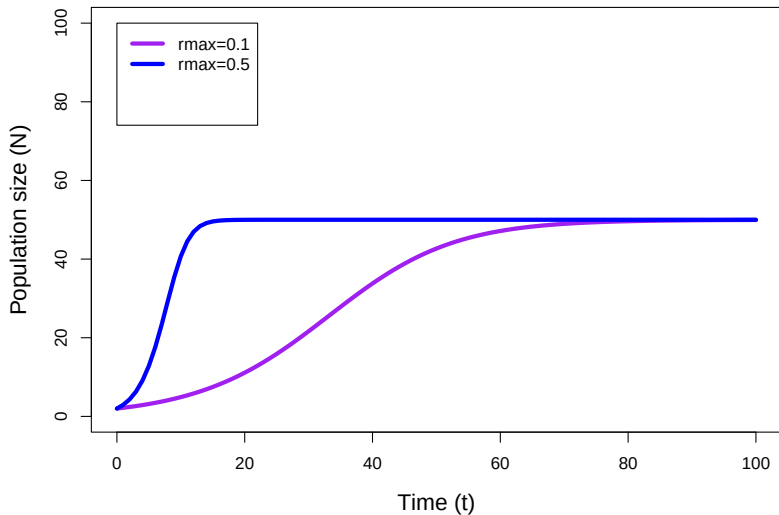
GROWTH ($\Delta_t = N_{t+1} - N_t$) AS A FUNCTION OF N



WHAT HAPPENS WHEN WE CHANGE r_{max} ?

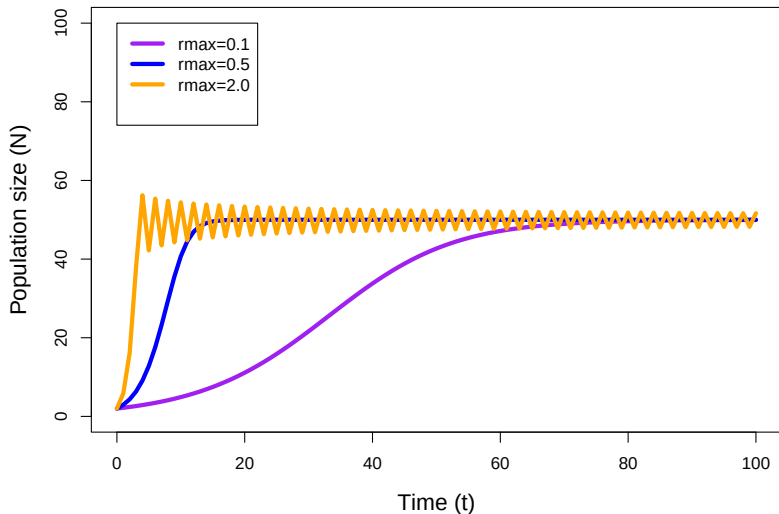


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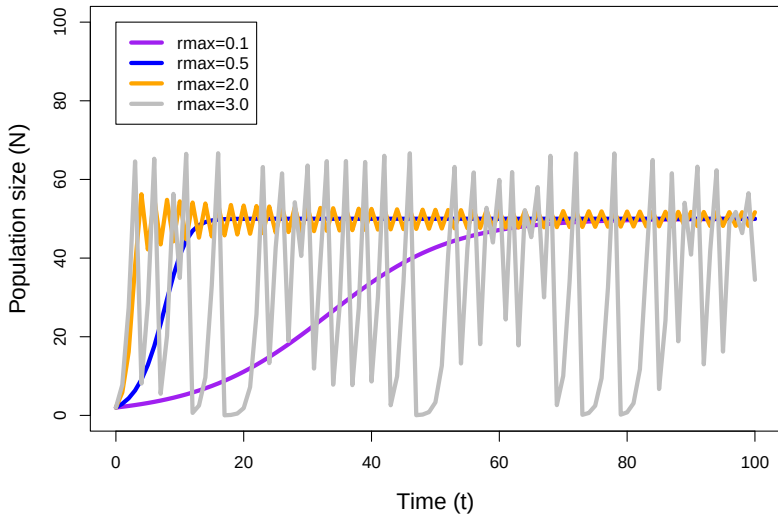
WHAT HAPPENS WHEN WE CHANGE r_{max} ?

Damped oscillation



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Chaos



OVERCOMPENSATION

Density-dependent response in which populations over- or under-shoot carrying capacity rather than approach it gradually

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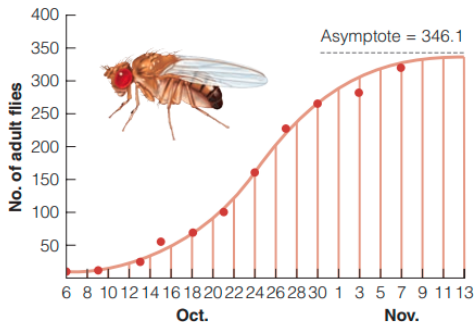
Highly variable deterministic dynamics that are extremely sensitive to small changes in parameters

- K and r_{max} are constant
- No sex or age effects or other sources of individual heterogeneity
- No time lags
- No stochasticity

DOES LOGISTIC GROWTH OCCUR IN NATURE?

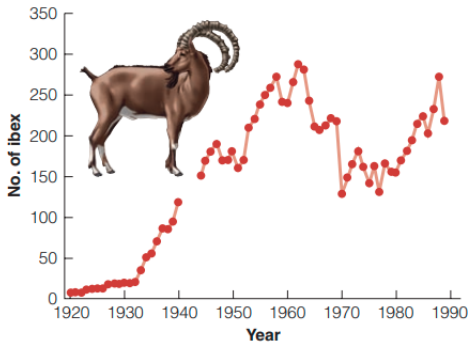
DOES LOGISTIC GROWTH OCCUR IN NATURE?

Fruit fly (*Drosophila melanogaster*) in the lab



DOES LOGISTIC GROWTH OCCUR IN NATURE?

Ibex (*Capra ibex*) in Switzerland.



Data from Sæther et al. (2002). Figure from Krebs (2014)

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Assignment

Read pages 32–36 in Conroy and Carroll